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Paleomagnetic evidence for Cenozoic clockwise rotation of the external Albanides

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Abstract

A paleomagnetic study of 750 samples obtained from 55 late Eocene to middle Pliocene sedimentary sites demonstrates a clockwise rotation of about 45° of the external zones of Albania. This rotation occurred either in two phases of roughly equal amplitude during the early–middle Miocene and Plio-Pleistocene or in a single phase with a clear acceleration of the movement during the Plio-Pleistocene. The good agreement of these results with those from southern Albania and northwestern Greece indicates that a large geographical area is involved in the rotation. The active 'Aegean' compressive front thus extends much farther north than hitherto believed and the rotational pattern is not significantly distorted by the tectonic discontinuities recorded by geological studies. The external zones of the Albano-Hellenic Belt appear to have rotated virtually as a single entity from the Peloponnesos to northern Albania over a deep decollement level probably involving the basement itself.

1. Introduction

The Albano-Hellenides Belt is an important feature of the Mediterranean Alpine belt which has mainly resulted from convergence between the African and European plates. Field studies have recorded the perfect continuity of the Albanian and Hellenic geological units, with a very regular trend of the structures (N150°) [1]. North of Albania the general trend changes to the typical Dinaric trend (N120°). This 'deflection' of the structures coincides with a major crustal discontinuity located on the northern edge of Albania, the so-called Scutari–Pec transverse zone. This fault zone, which also characterized by the disappearance of the karst zone (or Albanian Alps zone) to the south and by the 100 km westward displacement of the Albanian ophiolitic nappe (Mirdita nappe) with respect to its Dinaric equivalent has been described as a paleotransform fault that was active during the Mesozoic [2]. Discontinuities of smaller amplitude also exist in the Albano-Hellenic chain. The largest, the Vlore-Elbasan-Diber transverse zone, is located in central Albania. Although it is very active seismically [3] it does not correspond to any significant change in the orientation of the structures.

The Albano-Hellenic orogen is an imbricated stack of nappes thrust southwestward over the Adriatic and northern Ionian foreland. North of

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Kephallinia a continental type collision exists in the northern Hellenides, Albanides and Dinarides. The lithosphere changes southward into a thin continental one, and even into an oceanic one beneath the Ionian abyssal plain [4]. South of Kephallinia, the subduction of this lithosphere beneath the Hellenic arc is associated with a well-defined trench.

In the last decade paleomagnetic studies have shown rotations about a vertical axis that have accompanied the complex geodynamical evolution of the central Mediterranean Alpine belt. The most recent synthesis of the available data reveals that the present curvature of the Hellenic arc was acquired by opposite post-early Miocene rotations at the two terminations of the arc [5]. In particular, the western part of the arc has undergone a 45–50° Cenozoic clockwise rotation, which occurred in two distinct phases of about equal amplitude. The most recent has been recorded in the Ionian and in the pre-Apulian zones of the Ionian Islands, on both sides of the Kephallinia transform zone. This started about 5 Ma ago and continued since then at an average rate of about 5°/Ma [6]. This movement was preceded by a period of about 7 Ma during which no significant rotation occurred. An earlier phase of 20–25° clockwise rotation has been inferred from the results obtained from the Ionian zone in mainland Greece [7]. The period of activity of this rotation has not accurately been determined between the late Oligocene and late Miocene but has been related by us to the major Langhian tectonic phase. No other significant Tertiary rotation seems to have occurred prior to this [7,8].

Recently, new paleomagnetic data have been obtained north of the Greek border in Albania. Mauritsch et al. [9] have reported large Neogene clockwise rotations in the internal zones of southern Albania. In the Ionian zone, a mean paleodeclination of about 45°, similar to that observed in northwestern Greece, has been obtained from Eocene and Oligocene formations [10]. North of this area there is a wide zone that remains unexplored paleomagnetically; the closest area studied is the Kvarner Islands and vicinity, where the post-Cretaceous data deviate westerly, consistent with the African apparent polar wander path [11]. Therefore, although the investigated time periods are not the same, the rotational pattern in the northern Dinarides appears to be clearly different from that in southern Albania and northwestern Greece. Additional paleomagnetic studies are thus needed to obtain data to fill the gap in the external Dinarides and Albanides and to determine the possible decoupling role played by the Scutari–Pec transverse zone.

In this paper we report paleomagnetic results obtained from upper Eocene–middle Pliocene sedimentary formations in central Albania. Together with the results already available (discussed above) and within the framework of known geological data our data allow a fairly complete description of the Tertiary geodynamic evolution of the external Albano-Hellenides.

2. Outline of the tectonics and sampling

The Albanides orogenic belt is divided into different paleogeographic units that have been folded and thrust during the Alpine orogenesis [12]. The Sazani, Ionian and Kruja zones are the 'isopic' zones constituting the external Albanides (Fig. 1). The Kruja zone (equivalent to the Greek Gavrovo Zone) is composed of Cretaceous–Eocene platform limestones overlain by Oligocene–Aquitanian flysch. The Kruja zone was thrust over the Ionian zone during the early Miocene (probably late Aquitanian). This zone is limited geographically to a narrow band parallel to the chain in which the orientation of the structures is a perfect N150° (except in northern Albania where two calcareous ridges have a Dinaric trend of N120°).

The pelagic Ionian zone covers the major part of southern Albania. It is cut by minor thrusts striking N150°, most probably over a Triassic evaporite decollement level. The Ionian zone has been affected by two major tectonic phases. The first occurred during the middle Miocene (Langhian) and the second occurred at the Mio-Pliocene boundary, during which the Ionian zone was thrust onto the Sazani (pre-Apulian) zone, which forms a narrow peninsula in southwestern Albania.

In central Albania, the structural axes of the Ionian zone disappear below the Upper Miocene–Upper Pliocene detrital sediments of the 'pre-Adriatic Depression' (Fig. 1). Although no borehole could reach the limestone substratum, this large basin is assumed to cover the Ionian zone. Recent and probably still active tectonic deformation has affected the pre-Adriatic Depression, as shown by the strongly tilted early–late Pliocene deposits outcropping along the Adriatic coast. North of Tirana the Tortonian molasse of the pre-Adriatic Depression unconformably overlies the most external thrust sheets of the Kruja zone (Fig. 1). This molasse is locally deformed, suggesting out-of-sequence thrust reactivations within the Kruja zone.

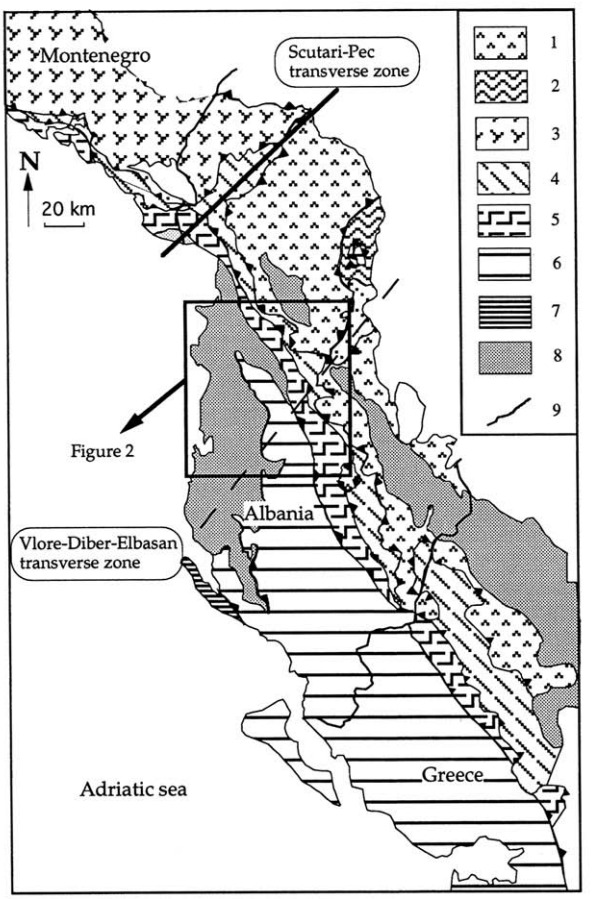


Fig. 1. Structural map of Albania and surrounding regions, showing the location of the studied area and the regional trend of the Albano-Hellenic belt. *Internal zones*: *1* = Mirdita–sub-Pelagonian zone; *2* = Korab zone; *3* = Karst–Albanian Alps; *4* = Budva–Krasta Cukali–Pindus zone. *External zones*: *5* = Dalmate–Kruja–Gavrovo zone; *6* = Ionian zone; *7* = Sazani–pre-Apulian zone; *8* = Albano-Thessalian and pre-Adriatic depressions. *9* = Political borders.

The NE–SW Vlore–Diber–Elbasan discontinuity which cuts the Albanian structures is marked by Permo-Triassic evaporite domes and by an elongate structural high along which the external zones are visible within the internal domain [13]. These tectonic windows are strong evidence for the complete allochthony of the Albanian units. Neither changes in the structural axis directions of the Ionian and Kruja zones nor strike-slip shifts of adjacent tectonic units are observed on either side of this seismically active zone (Fig. 2).

A total of 74 paleomagnetic sites have been sampled in central Albania (Fig. 2), 13 sites in fine-grained platform limestones from the Kruja zone and 61 sites in flysch and molasse sections of both the Kruja zone and the pre-Adriatic Depression. In these detrital formations blue fine-grained marls in fresh outcrops were preferentially selected because previous studies have shown that this type sediment was deposited in calm water and suffer no strong weathering [6,7]. By the term 'site' we mean an outcrop that is several tens of metres wide and several metres high. At each site we collected ten to fifteen cores that were well distributed both horizontally and vertically in order to average out the secular variation of the geomagnetic field and to obtain a representative statistical mean direction. Dating of the sites was obtained by studying associations of foraminifera in the limestones and of nannoplankton in the clay sections. This biostratigraphic study generally confirmed the ages given by the 1:200,000 geological map of Albania [14], with the exception of some as discussed below.

In the Kruja zone both upper Eocene (Priabonian) shelf limestones (thirteen sites) and Oligocene flysch (twelve sites) were sampled in the Dajti–Kruja ridge, east and northeast of Tirana. The strike of this monocline is extremely regular and coincides with the regional trend (N150°) throughout central Albania. The sites are distributed in two of the Kruja tectonic slices.

South of Elbasan seven sites were sampled in the flysch and molasse close to Gramsh (Fig. 2) along minor thrust sheets striking N–S. These formations are shown as Oligocene on the geological map, but three sites (AL97, AL98, AL100) were found to belong to the NN9–NN12 zones and are thus of Upper Miocene (Serravallian–Messinian) age. The deformed clays outcropping in this zone

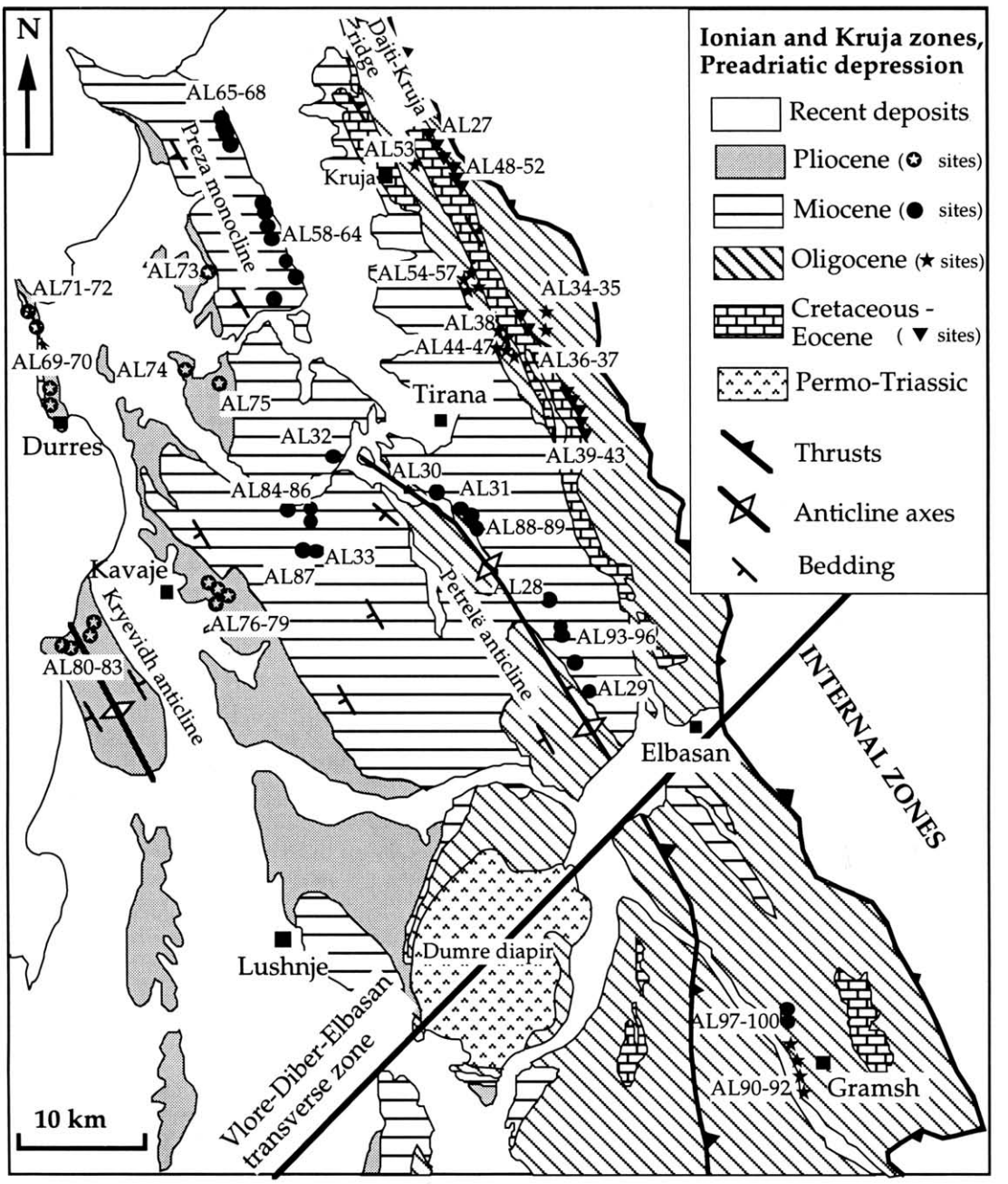


Fig. 2. Geological map of the studied area and location of paleomagnetic sampling sites.

confirm that the Kruja zone has been deformed since the early Miocene.

In the pre-Adriatic Depression we sampled 27 sites in flysch and molasse of Burdigalian–Tortonian age (Fig. 2). These sites are distributed partly along the eastern limb of a wide regional anticline south of Tirana (Petrelë anticline), and partly in the Preza monocline, which overthrusts to the east the younger deposits of the Kruja–Tirana plain. This monocline is the northern continuity of the Petrelë anticline and the total axial length of this major structure is about 70 km. New biostratigraphic determinations place these formations in the NN9–NN12 zones (Serravallian–Messinian). They are thus younger than previously believed [14].

Finally, fifteen sites were sampled in early Pliocene molasse in three different structures in the western part of the pre-Adriatic Depression. Seven are located on the western side of the Petrelë–Preza structure, four in the hinge zone of the Kryevidh anticline, close to the coast, and four along the Durres backthrust in strongly dipping (or even overturned) layers.

All cores were drilled using standard techniques and oriented using both sun and magnetic compasses. More than 1000 cores were obtained from the 74 sites, each core giving two to five samples for paleomagnetic analyses.

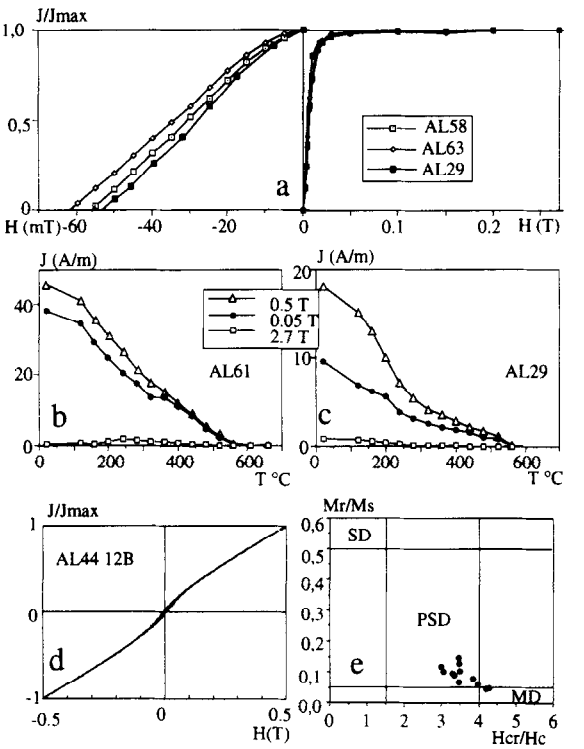


Fig. 3. Magnetic properties of the marly samples. (a) Stepwise acquisition of the isothermal remanent magnetization (IRM) and determination of the remanent coercivity (H_{cr}). (b) Thermal demagnetization of a three-component IRM produced by magnetizing the samples in fields of 2.7, 0.5 and 0.05 T successively along the three different sample axes. (c) Thermomagnetic curve obtained by Curie-balance analysis. (d) Hysteresis loop. (e) M_r/M_s vs H_{cr}/H_c diagram [16].

3. Laboratory measurements

3.1. Magnetic properties

In order to identify the magnetic carriers, coercivity and thermal spectra were investigated for at least one sample per site. The studied formations can be separated into two groups depending on facies—detrital (marls and clays) and calcareous neritic limestones. The results obtained from these two groups of samples are shown in Figs. 3 and 4.

The first experiment that was conducted was the stepwise acquisition of isothermal remanent magnetization (IRM) followed by the determination of the coercivity of remanence (H_{cr}) applying a progressively increasing reverse field. The detrital formations exhibit an almost complete saturation of the IRM at relatively low fields (about 0.2–0.3 T) and low values of H_{cr} (between 35 and 65 mT) (Fig. 3a). These values are consistent with the presence of magnetite but cannot exclude the possible presence of iron sulphides, which are characterized by similar values of these parameters.

In order to determine more precisely the magnetic minerals, we thermally demagnetized step by step three mutually perpendicular IRMs given at fields of 2.7, 0.5 and 0.05 T along the three sample axes [15]. The absence of a hard component (2.7 T) is confirmed. Moreover, in about 85% of the samples both the medium (0.5 T) and soft (0.05 T) components are very progressively

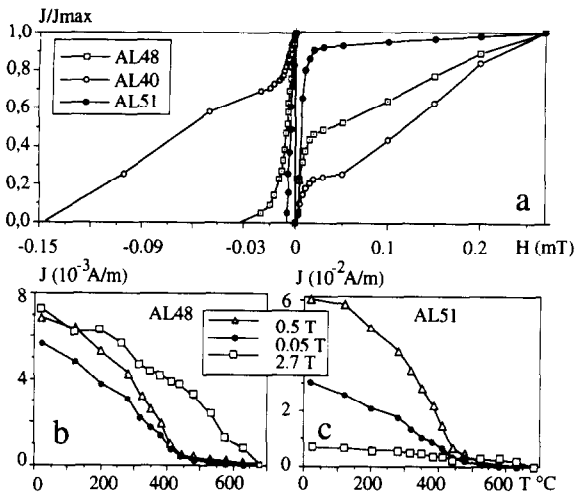


Fig. 4. Typical magnetic properties of representative calcareous samples. (a) Stepwise acquisition of the isothermal remanent magnetization (IRM) and determination of the remanent coercivity. (b) and (c) Thermal demagnetization of a three-component IRM for two samples showing different behaviour.

removed up to 560–580°C (Fig. 3b). Relatively pure magnetite is thus the main magnetic carrier in these samples. In the other 15% of the samples we observe about the same behaviour, except for a change in the slope at about 300°C indicating that another magnetic carrier, possibly an iron sulphide (greigite?), is associated with the magnetite (Fig. 3c).

Finally, hysteresis loops were obtained from small chips of bulk rocks (Fig. 3d). After removal of the paramagnetic contribution due to the clay content, identified by the positive slope at high fields, saturation magnetization (M_s), saturation remanence (M_{rs}) and coercivity (H_c) were determined. The coercivity of remanence (H_{cr}) was also obtained, by applying a progressively increasing back field after saturation. The ratios of these different parameters indicate that the magnetic properties of the marly samples are mainly carried by magnetite in the pseudo-single domain range [16] (Fig. 3e).

The limestones show a more complex mineralogy, with sometimes the coexistence of low-coercivity minerals saturating at 0.2–0.3 T, and high-coercivity minerals that are not yet saturated at the maximum field reached of 2.7 T (Fig. 4a). The coercivities of remanence (H_{cr}) range between 70 and 1500 mT, depending on the relative percentage of low- and high-coercivity minerals (Fig. 4a).

The three-axis experiment confirms that the contribution of the hard component is highly variable and shows that this component is progressively removed up to 680°C, suggesting hematite as the main magnetic carrier (Fig. 4b). This may be surprising considering the white colour of the samples. The low and intermediate coercivity components are almost completely demagnetized at 400–450°C (Figs. 4b and c), illustrating magnetite contributions with a moderate Ti-content. The limestones were too weakly magnetized to allow hysteresis loops to be obtained.

3.2. Stability of the remanent magnetization

Measurement of remanent magnetization was carried out using a three-axis cryogenic magnetometer in a shielded room. Generally the marly formations have a much higher intensity (5×10^{-3} A/m) than the limestones (5×10^{-5} A/m). Seven out of the thirteen calcareous sites from the Kruja zone were too weakly magnetized ($< 10^{-5}$ A/m) to be accurately measured.

One pilot sample per measurable site was first demagnetized in close steps (20–40°C) in order to select the correct heating procedure for the other samples, all of which were then systematically demagnetized in eight to fourteen steps from room temperature to the limit of reproducible results. In a general way we have observed two different types of demagnetization behaviour. In the first case (55%) after removal of a viscous component at 120–280°C a single stable component of magnetization could be isolated (Fig. 5). The intensity is reduced to about one tenth of its initial value around 480–500°C in the limestones (Fig. 5a) and 520–600°C in the marls (Fig. 5b). In 30% of the samples two components of magnetization, with overlapping blocking temperature spectra, are present in the samples. In these cases the great circles fitting the two directions have been calculated. Eight marly sites were unstable during demagnetization and yielded no reliable results.

Finally, tightly grouped within-site directions are obtained, as illustrated in Fig. 6. More quantitatively, the mean directions of magnetization for each site have been calculated using Fisherian statistics [17] when only stable directions are observed and using the McFadden and McElhinny [18] method when great circles are present. The data obtained from the 59 reliable sites are reported in Table 1 together with the mean regional directions for each period. The confidence angle α_{95} varies for each site between 1.5 and 15°, with the greater majority between 5 and 7° indicating that the mean directions are accurately defined. The same accuracy is obtained for the regional mean directions which are calculated for each period (Table 1). Four Oligocene mean directions diverged by more than 2σ from the mean regional direction mainly because of very shallow inclinations (Table 1). These directions were not included in the final statistics. The directions yielded by the six sites sampled in the Gramsh structure are consistent with the mean directions after they are placed in a regional framework by rotating the local fold axis strike along the regional trend, similar to what has been previously done in Epirus [8] (Table 1).

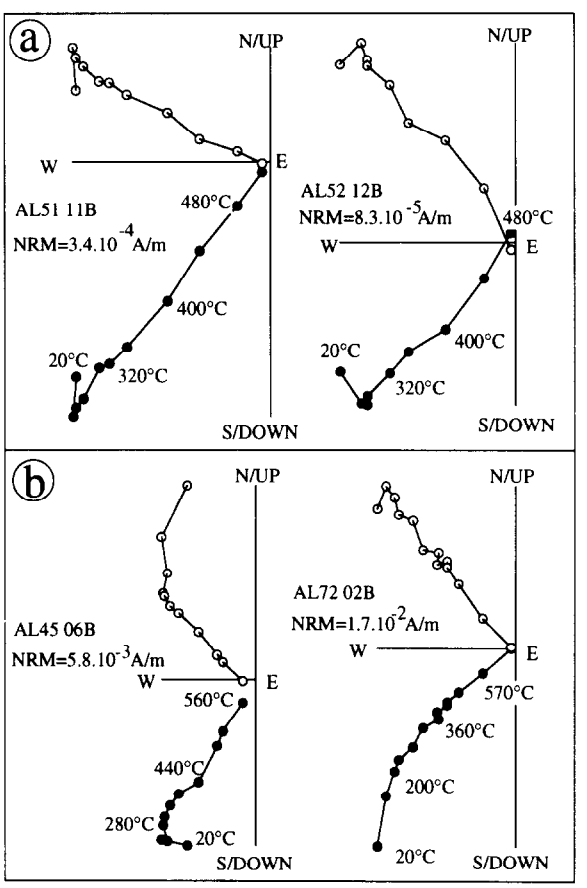


Fig. 5. Typical thermal demagnetization diagrams of the calcareous (a) and the marly (b) samples. ○/● = Projections onto the vertical/horizontal plane.

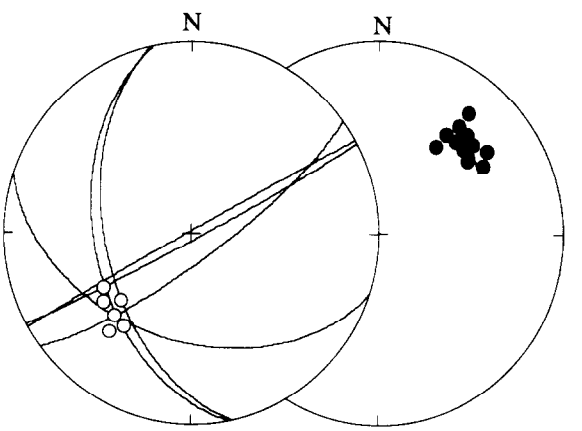


Fig. 6. Typical results cluster obtained from sites characterized by only stable directions or by both stable directions and great circles (equal-area projection). Solid lines and circles (○) = upper hemisphere projection; ● = lower hemisphere projection.

Upper Miocene sites show normal and reverse polarities with virtually antipodal mean directions after tectonic correction (positive reversal test of the type R_c, [19]). This is good evidence of magnetic stability and satisfactory cleaning (Fig. 7). In addition, regional positive tilt tests could be obtained for almost every period. For example, in Fig. 7 the Upper Miocene directions are clearly better grouped after than before bedding correction (at the 1% confidence level [20]); this also applies to the Oligocene and Lower–middle Pliocene (Table 2). All the upper Eocene sites from the Kruja zone show the same bedding attitude as they were sampled in the same tectonic slice. Thus no tilt test could be applied to

Table 1
Paleomagnetic directions from the external Albanides

Site	$(n_1,n_2)/n_3$	$D_{bc}(°)$	$I_{bc}(°)$	$D_{ac}(°)$	$I_{ac}(°)$	K_{ac}	$\alpha_{95ac}(°)$
Upper Eocene sites (Kruja Zone)							
AL27	(7,2)/10	56	-79	222	-41	146	4.3
AL48	(4,7)/11	79	-65	232	-52	93	5.0
AL49	(8,4)/12	116	-82	228.5	-35	23	9.4
AL50	(3,7)/13	82	-81	235	-33	185	3.8
AL51	(7,3)/12	61	-82.5	228	-37	55	6.7
AL52	(6,6)/12	77	-76	228	-42	102	4.4
Mean direction	bc: n/N = 6/13		D = 77.5°		I = -78°	K = 115	α_{95} = 6.3°
	ac: n/N = 6/13		D = 229°		I = -40°	K = 114	α_{95} = 6.3°
Oligocene sites (Kruja zone)							
AL34	(4,4)/10	340	-87	228.5	-49	38	9.5
AL35	(4,4)/8	344	-76	234.5	-38	26	11.5
AL38*	(15,0)/15	227	-26	228	4	54	6.0
AL44	(13,0)/13	228	-51	230	-39	53	5.7
AL45	(13,0)/15	212	-32	212	-32	67	5.1
AL46*	(12,0)/14	225	-10	224	-13	65	5.4
AL47*	(12,0)/15	233	-10	234	8	39	7.4
AL53	(10,1)/11	232	-68	243	-42	29	8.6
AL54	(0,8)/14	209	-67	229	-34	1303	1.5
AL55	(12,1)/15	185	-53	209	-28.5	122	3.8
AL56	(7,5)/15	166	-68	211	-43	65	5.5
AL57	(10,2)/14	82	-80	223	-43	52	6.1
AL90°	(10,1)/12	253	-28	234 (253)	-47	43	7.1
AL91°*	(9,0)/12	239	-61	224 (243)	-16	120	4.7
AL92°	(6,2)/11	240	-37	221 (240)	-53	27	11
AL99°	(4,3)/11	237	-61	232 (251)	-42	22	14
Mean direction	bc: n/N = 12/17		D = 224°		I = -65°	K = 9.2	α_{95} = 15.1°
	ac: n/N = 12/17		D = 225°		I = -41°	K = 57	α_{95} = 5.8°
Upper Miocene sites (Preadriatic depression)							
AL30	(7,2)/12	207	38	209	-25	92	5.5
AL31	(10,4)/14	56	-42.5	210	-56	94	4.2
AL32	(0,11)/14	36	69	42.5	55	23	8.5
AL33	(7,5)/12	49	53	69	48	72	5.3
AL58	(12,4)/16	40	11	9	54	19	8.6
AL59	(0,13)/13	48	33	28	53.5	19	9.6
AL60	(5,6)/13	48	17	48	52	42	7.3
AL61	(12,1)/13	37	21	23	51	41	6.6
AL62	(11,2)/13	38	17	39	46	29	7.9
AL63	(8,6)/14	24	18	15	43	49	5.8
AL64	(4,3)/14	216	-16	206	-41.5	17	15.5
AL66	(8,3)/12	47	6	42.5	33	23	9.8
AL67	(12,1)/13	49	9	44	34	107	4.0
AL68	(10,3)/13	35	12	29	32	41	6.6
AL84	(6,6)/12	58.5	47	55	49	104	3.5

Table 1 (continued)

AL85	(4,6)/11	57	23	39	52	36	8.5
AL86	(0,10)/13	54	26.5	40	47	39	7.8
AL87	(12,0)/12	50	35	45	42	64	5.5
AL89	(5,1)/13	239	20	11.5	27.5	77	7.8
AL93	(10,0)/11	335.5	61	44	30	78	5.5
AL94	(8,0)/12	313	72	32	49	37	9.2
AL95	(7,4)/12	313.5	66	34	33	61	6.0
Upper Miocene sites (Kruja zone)							
AL98°	(10,0)/13	9	66	34 (53)	37	45	7.3
AL100°	(2,3)/12	355	68	44 (63)	36	54	12.0
Mean direction	bc: n/N = 24/29		D = 35°	I = 46.5°	K = 4	α_{95} = 18.2°	
	ac: n/N = 24/29		D = 35°	I = 44°	K = 35	α_{95} = 5.0°	
Lower-Middle Pliocene sites (Preadriatic Depression)							
AL69	(4,2)/13	150	-57	215	-34	31	12.0
AL70	(6,8)/15	128	-42	192	-55	66	5.0
AL71	(9,6)/15	105	1	180	-55	116	3.6
AL72	(1,2)/13	93	19	234	-58	44	6.4
AL74	(5,5)/10	205	-34	197	-44	33	8.8
AL76	(5,5)/13	190	-7	185	-31	88	5.3
AL77	(7,4)/11	227	-20	230	-53	158	3.7
AL78	(3,7)/12	212	-10	208	-41	29	9.5
AL79	(1,12)/13	205	-21	196	-50	14	12.0
AL80	(7,5)/13	206	-39	207	-41	32	8.0
AL81	(9,3)/12	203	-49	210	-49	45	6.6
AL82	(5,8)/13	214	-68	224	-61	48	6.2
AL83	(5,8)/13	191	-52	200	-43	46	6.3
Mean direction	bc: n/N = 13/15		D = 182°	I = -38°	K = 3.4	α_{95} = 26°	
	ac: n/N = 13/15		D = 205°	I = -48°	K = 33.5	α_{95} = 7.2°	

$(n_1,n_2)/n_3$ = (number of stable directions.number of great circles)/total number of studied samples at the site. *D*eclination and *I*nclination are reported before (bc) and after (ac) tectonic correction. For each period, the mean regional direction is also calculated before and after tectonic correction. n/N = number of sites taken into account in the statistical calculation (see text)/total number of studied sites; *sites rejected at the 2σ level; °sites corrected for the local fold axis orientation (see text) (the declinations before this correction are between parentheses).

Table 2
Fold tests [20] for the different periods investigated in Albania

Age	Number of sites	K2/K1	F(1%)	Result
U. Eocene	6	1.0	4.8	not significant
Oligocene	12	6.2	2.8	positive
U. Miocene	24	8.7	2.0	positive
L. M. Pliocene	13	9.8	2.7	positive

$K2/K1$ = Dispersion parameters after and before tectonic correction respectively.

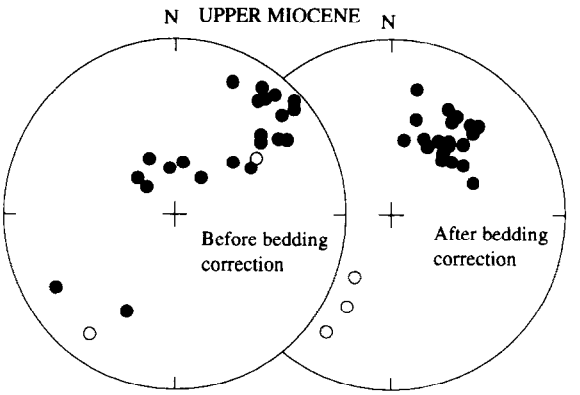


Fig. 7. Equal-area projections of the mean directions of magnetization obtained from Upper Miocene sites. The scatter of the data decreases significantly (at the 1% significance level) after tilt correction. ●/○ = downward/upward directions.

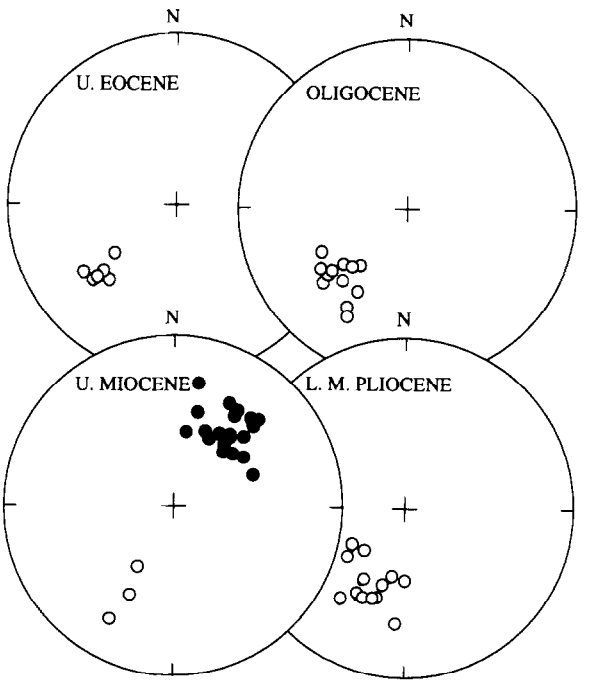


Fig. 8. Equal-area projections of the site-mean directions after structural tilt correction obtained for each period from the external Albanides. Same symbols as in Fig. 7.

these sites. However, their paleomagnetic directions before tectonic correction are completely different from the present-day field direction and completely aberrant ($I = -78°$, Table 1). This rules out the possibility of a post-tilting magnetic overprint. Moreover, these Eocene directions after tectonic correction are perfectly similar to the coeval ones obtained previously in southern Albania [10].

In summary, the laboratory analyses and the complementary field tests provide evidence that the primary component of magnetization was successfully isolated at 55 sites and that most of the investigated formations yield paleomagnetic di-

Table 3
Rotation and flattening of the Albano-Greek margin with respect to Africa

Region	Age of the formations	N	R	F	Ref.
Centr. Albania	U. Eocene	6	44°±7°	11°±6°	this paper
"	Oligocene	12	38°±7°	13°±5°	this paper
"	U. Miocene	24	30°±6°	12°±4°	this paper
"	L. M. Pliocene	13	23°±9°	10°±7°	this paper
South. Albania	Eocene	11	40°±5°	10°±5°	[10]
Epire	Eocene	51	37°±5°	6°±4°	[8]
Epire + Akarn.	Oligocene	29	51°±5°	14°±4°	[7]
Ionian Islands	U. Miocene	4	22°±5°	11°±4°	[6]
"	Plio-Pleistocene	12	variable from 25° to 0°		[6]

N = Number of sites; R and F = rotation and flattening parameters calculated using the equations given by Beck [22] and Demarest [23]. The data have been related to the African APWP proposed by Besse and Courtillot [21].

rections which can be interpreted in terms of a geodynamic evolution.

4. Results and discussion

The mean regional paleomagnetic directions calculated for different age intervals for sites yielding reliable results are characterised by significant easterly declinations (Fig. 8 and Table 1). To interpret these results in terms of rotation, in the absence of available Cenozoic poles for the Adriatic foreland, we have related our data to the most recent African apparent polar wander path (APWP) published by Besse and Courtillot [21].

In Table 3 we have reported the rotation (R) and the flattening (F) parameters calculated for each period [22,23]. To make comparison easier, we have also reported in this table the parameters for coeval formations in southern Albania and northwestern Greece calculated from previous results which had been related to other APWPs at the time.

On average, inclinations appear to be 10° shallower than expected. Shallow inclinations have already been reported from Cenozoic lava flows and limestones in Greece [5] and generally from all over the Tethyan realm [24]. Possible geomagnetic [24,25] or tectonic [26] explanations have been suggested. However, so far there is no clear consensus among the different authors. For the marly formations studied here it may be possible that a significant part of this 'inclination error' is due to compaction occurring just after deposition [6]. So it appears unreasonable to interpret the flattening parameter in terms of northward drift.

As far as the rotations are concerned, the 45° clockwise rotation observed in the Eocene limestones is similar to the value obtained from Oligocene formations, implying that the rotation is younger than Oligocene. As shown in Fig. 9, the clockwise rotation does not seem to have occurred at a regular rate since the Oligocene. The early Pliocene results (23 ± 9°) clearly show that part of the rotation occurred very rapidly in the last 5 Ma. The timing of the Miocene phase is

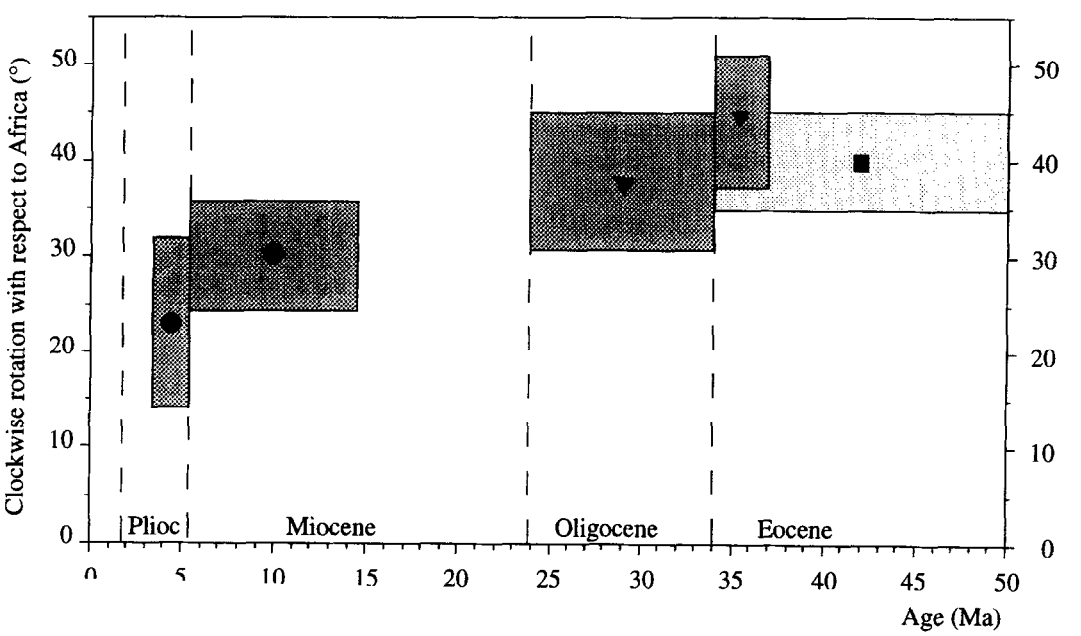


Fig. 9. Clockwise rotation of the external Albanides with respect to Africa vs time during the Cenozoic period. The numerical data are in Table 3. *Tectonic units*: ● = Pre-Adriatic Depression; ▼ = Kruja zone; ■ = Ionian zone. Dark and light grey boxes correspond to the results reported in this paper and in Speranza et al. [10] respectively.

not so clear and two scenaros are possible. First, we observe that the values of rotation obtained from the late Miocene and the early Pliocene periods are not significantly different. Thus this time interval may correspond to a period of no rotation between a 15–20° early–middle Miocene phase and the recent one. In the second scenario the post-early Pliocene rotation which is characterized by a minimum rate of 5°/Ma is just an acceleration of a first slower rotation which started during the early Miocene without any intermediate period of nil rotation. The lack of reliable Lower–Middle Miocene data does not allow us to better constrain the timing of the rotation or to propose a clear-cut interpretation.

This rotational pattern is similar to that previously reported to the south in the pre-Apulian and Ionian zones of the external Hellenides [5–8]. The post-Oligocene 45° clockwise rotation characterized by a very rapid phase during the last 5 m.y. is thus now well established from the Peloponnesus to central–northern Albania.

This first indicates that the Vlore–Diber–Elbasan tectonic discontinuity did not play a significant role in the post-Eocene rotational behaviour of the external Albanides. Such rotation is difficult to explain in the continental collision setting existing in northwestern Greece and Albania. So one may be tempted to consider that only the cover may be involved. However, the geographical and structural distribution of the reliable paleomagnetic sites indicates that this rotation certainly does not result from differential movements of the nappes. For example, the Eocene paleomagnetic data obtained from the Kruja zone in central Albania are identical to the coeval data obtained from the Ionian zone in southern Albania [10] and in northwestern Greece [8], indicating that the thrusting of the Kruja zone over the Ionian zone, which occurred during the early Miocene and underwent reactivation during the late Miocene, has not introduced differential rotation. The same observation had already been made for the Ionian and pre-Apulian zones in the Ionian Islands [6]. At a larger scale, post-Eocene rotations of virtually the same amplitude as that observed in the external zones have also been obtained from the piggy-back Albano-Thessalian basin, both in Greece [5] and in Albania [9], showing that the Krasta Cukali–Pindus thrust over the Kruja–Gavrovo and Ionian zone has not contributed significantly to the observed rotation. This implies that the rotation is accommodated by a much deeper discontinuity than the Triassic evaporite horizons that represent the main decollement level of the external zone nappes. If this interpretation is correct the Albano-Greek lower crust has been largely involved over the last 15–20 Ma in the overthrusting of the Adriatic–Ionian foreland, as recently suggested for the southern part of the Aegean arc by Jolivet et al. [27].

By placing the pivot point immediately to the north of the studied region, at the latitude of the Scutari–Pec transverse zone, and assuming in a very rough first approximation a perfectly rigid rotation, we can estimate the horizontal displacement resulting from a 45° rotation at about 170 km at the Albanian–Greek border and at about 350 km at the latitude of the Gulf of Corinth. So far, no surface expression of any important crustal discontinuity has been recorded in the Adriatic foreland west of Albania, and the question of the westward accommodation of the horizontal displacement due to the observed rotation is still open.

The remarkable consistency of the paleomagnetic data from the Peloponnesus to northern Albania is unexpected when one considers the geophysical models developed so far for the Aegean region. Most of these models [28,29] assume that the only mobile regions which can be affected by a rotation are located south of the Kephallinia fault, often described as the boundary between the active subduction of the Ionian basin along the Hellenic trench and the continental collision between the Adriatic 'plate' and northwestern Greece and Albania [4,28,30]. Contrary to these models, paleomagnetic evidence supports the hypothesis that no major differences in the rotational movements have occurred from the Peloponnesus to northern Albania. Discontinuities such as a possible boundary between subduction and continental collision, or such as the Vlore–Diber–Elbasan transverse zone, appear to have played a very minor role in the rotation.

This suggests that the Hellenic arc geodynamic system, which is still undergoing active movement, extends further north than previously believed. This would accord with recent re-evaluations of the focal mechanisms of earthquakes that occurred on the Adriatic coast of Albania, which show thrust mechanisms on planes parallel to the coast [31], similar to the pattern observed in the Ionian Sea.

The northern limit of the region involved in the clockwise rotations lies somewhere between central–northern Albania and the Kravner Islands. Our data show that this limit cannot be located south of the Scutari–Pec transverse zone, which corresponds to the boundary between the Dinarides and the Albano-Hellenides and which has already been suggested as the possible decoupling zone [5,32]. In order to definitively locate this boundary additional paleomagnetic data are now needed along the Dinaric margin.

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